Lab Report Problems

Sl. No.	Problems

1.	The SEU library system stores book titles entered by users. Since users may type titles using mixed case letters, numbers, and spaces, the system must standardize and analyze the text before saving it to the database. The SEU library uses an old 8086 simulated computer for this task. You are hired as a system programmer to develop the required program.
	
	Input:
	
	a. A single line string (max 30 characters)
	b. The string may contain-
	    1. Uppercase letters (A-Z)
	    2. Lowercase letters (a-z)
	    3. Digits (0-9)
	    4. Spaces
	
	Output:
	
	a. Print the formatted title (all uppercase)
	b. Print the list of vowels
	c. Print the list of consonants
	d. Print the total vowel count
	
	Sample Test Cases:
	
	Input:
	
	Book Title: Microprocessors and Interfacing 103
	
	Output:
	
	Formatted Title: MICROPROCESSORS AND INTERFACING 103
	
	Vowels:
	
	I O O E O A I E A I
	
	Consonants:
	
	M C R P R C S S R S N D N T R F C N G
	
	Total Vowels = 10

2.

In the SEU assembly programming lab, an 8086 oriented monitoring system is used to visualize CPU load levels during practical sessions. Each student program execution generates a load code (digits from 0 to 9), the system displays the load using star (*) patterns. However, the visualization logic depends on whether the load code is Odd or Even that ensures the program weight at the assembly level. You are assigned as an assembly programmer to implement this load logic.

Input:

Input load consists of one single digit decimal number ($0 \le N \le 9$).

Output:

a. Print whether the load is Odd or Even.
b. Print the load visualization of the number based on the following rules:

    1. If the number is Even, print decreasing star (*) up to N.
    2. If the number is Odd, print increasing star (*) up to N.
    3. If the number is Zero, print “All Are Stars Here”.

Sample Test Cases:

Input:
5

Output:

Odd Load Detected

```
*
**
***
****
*****
```

Input:
6

Output:

Even Load Detected

```
******
*****
****
***
**
*
```

	Input: 0 **Output:** All Are Stars Here
3.	A school is developing a simple math helper program using an 8086 simulated system. The program takes a number from the user and behaves differently based on whether the number is odd or even. Your task is to write an 8086 assembly language program to automate this process. **Input:** Input consists of one single digit decimal number ($0 \le N \le 9$). **Output:** a. Print whether the number is Odd or Even. b. Print the multiplication table of the number based on the following rules: 1. If the number is Even, print table from 1 to 10. 2. If the number is Odd, print table from 1 to 5. 3. If the number is Zero, print "All is Well". c. Each line of the table must be printed in the format, N x i = result. **Sample Test Cases:** **Input:** 4 **Output:** Even Number 4 x 1 = 4 4 x 2 = 8 4 x 3 = 12 4 x 4 = 16 4 x 5 = 20 4 x 6 = 24 4 x 7 = 28 4 x 8 = 32 4 x 9 = 36 4 x 10 = 40

	Input: 7 **Output:** Odd Number 7 x 1 = 7 7 x 2 = 14 7 x 3 = 21 7 x 4 = 28 7 x 5 = 35 **Input:** 0 **Output:** All is Well